

Week 16: Practice Assignment

Practice Assignment: Leveraging Deep Learning for Feature Extraction & Recommendation Systems

Note: *This is a practice assignment and will not be graded. Its purpose is to help you gain hands-on experience with Deep Belief Networks (DBNs) for feature extraction and Restricted Boltzmann Machines (RBMs) for recommendation systems. You'll explore two key deep learning use cases and reflect on their effectiveness in real-world contexts.*

Before you begin: Try solving the assignment on your own before looking at any solution sets. It's perfectly okay to make mistakes — the goal is to learn by doing, experimenting, and reflecting.

What this is about:

This assignment introduces you to two powerful applications of deep learning:

1. **Feature Extraction for Classification** using Deep Belief Networks (DBNs) on the MNIST dataset.
2. **Collaborative Filtering for Recommendations** using Restricted Boltzmann Machines (RBMs) to generate personalized user-item recommendations.

You'll learn to build and evaluate models that can learn hidden patterns in the data and apply them for classification and personalization tasks.

What you'll learn:

By completing this assignment, you'll gain experience in:

- Implementing Deep Belief Networks for hierarchical feature learning
- Training classifiers on deep-learned features for improved accuracy
- Building RBM-based collaborative filtering models
- Predicting user preferences and generating recommendations
- Evaluating model performance and tuning hyperparameters
- Presenting your methodology and insights effectively

Your Practice Tasks:

Part 1: Feature Extraction Using Deep Belief Networks (DBNs)

Dataset: MNIST Handwritten Digit Dataset

Steps:

- Preprocess the MNIST data (normalize pixel values, reshape as needed)
- Split the data into training and testing sets
- Build a Deep Belief Network (DBN) using stacked Restricted Boltzmann Machines (RBMs)
- Use the DBN to extract high-level features from the raw images
- Train a classifier (e.g., logistic regression or softmax) on the extracted features
- Evaluate the classification performance using accuracy on the test set

 **Reflection:** How did the DBN features affect the classifier's performance? □ Did feature extraction improve accuracy compared to raw pixel input?

Part 2: Recommendation System Using Restricted Boltzmann Machines (RBMs)

Dataset: Use a real-world or synthetic user-item interaction dataset (e.g., MovieLens or custom CSV)

Steps:

- Preprocess the data (e.g., binarize ratings: liked = 1, not rated = 0)
- Implement an **RBM-based collaborative filtering** model
- Train the model using a portion of the user-item matrix
- Tune hyperparameters such as learning rate, number of hidden units, and epochs
- Predict ratings or interactions for the test set
- Generate **Top-N personalized recommendations** for sample users

 **Reflection:** How well did the RBM capture user preferences? What challenges did you face in training or tuning the RBM? Would this approach scale well to large datasets?

Part 3 (Optional): Summarize in a Short Presentation

- Prepare a 3–5 slide presentation summarizing your work and insights:
- Overview of DBN and RBM models used
- Dataset descriptions and preprocessing steps
- Visuals of feature extraction and recommendations
- Performance metrics and lessons learned

Submission Checklist (Optional for Practice)

- ✓ MNIST data preprocessed and DBN built
- ✓ Feature extraction and classifier performance evaluated
- ✓ User-item dataset prepared and binarized
- ✓ RBM trained for collaborative filtering
- ✓ Recommendations generated and visualized
- ✓ Results and insights documented
- ✓ (Optional) Summary presentation created

Final Reflection

- What did this assignment teach you about unsupervised and generative deep learning methods?
- Which part—feature extraction or recommendation—was more intuitive or challenging?
- How could these methods be scaled or adapted to other domains like fraud detection or personalized learning?

Happy Experimenting!

This assignment gives you hands-on practice with two exciting deep learning architectures. By working through both classification and recommendation pipelines, you'll better understand how unsupervised learning and feature hierarchies power real-world intelligent systems.